Automatic Detection of Subsidence Funnels in Large-Scale SAR Interferograms Based on an Improved-YOLOv8 Model

Jing Guo, Zhengjia Zhang, Mengmeng Wang, Peifeng Ma, *Member, IEEE*, Wei Gao, and Xiuguo Liu, *Member, IEEE*

***Abstract*— Coal mining activities can induce ground subsidence, collapse, and even surface fissures, posing a severe threat to human safety. In this article, a novel method integrating interferometric synthetic aperture radar (InSAR) and convolutional neural networks (CNNs) was proposed for automated subsidence funnel identification. Initially, a hybrid InSAR dataset was constructed by combining real samples from mining areas obtained through interferometric processing with simulated samples synthesized using the probability integral method, Polin noise, and complex Gaussian white noise. Subsequently, on the basis of the YOLOv8 algorithm, the adaptive detection head dynamic head (Dyhead) based on attention mechanism and the regression box loss function Wise-intersection over union (WIoU) that can improve the problem of uneven sample difficulty were introduced, resulting in the proposed Improved-YOLOv8 model. Trained on the hybrid dataset, it significantly improved detection accuracy compared to five base models, achieving AP_{50}, AP_{75}, and AP_{50-95} of 92.0%, 60.0%, and 54.1% respectively. Further experiments and analyses indicate that the trained Improved-YOLOv8 model exhibits satisfactory applicability and accuracy for different surface types and other satellite datasets, and performs well in subsidence funnels detection task covering the entire Shanxi. Therefore, the proposed method shows significant application potential in determining the location distribution of subsidence funnels over wide areas, regularly updating data and monitoring geological disasters in mining areas.**

***Index Terms*— Deep learning, interferometric synthetic aperture radar (InSAR), simulated samples, subsidence funnel detection.**

I. Introduction

THE extraction of underground mineral resources generally causes significant ground collapses and the formation of subsidence funnels [1], posing potential threats to human lives and property safety, impeding the sustainable development of the mining economy [2]. Therefore, the precise and efficient localization of deformations plays a pivotal role in averting geological disasters and surveilling unauthorized mining operations [3]. Traditional geodetic measurements encompass a range of techniques, including precise leveling, total station measurement, the global navigation satellite system (GNSS), and various other on-site measurement methods [4]. While these methods boast high measurement accuracy, they require substantial human and material resources and suffer from limitations, such as low spatial resolution, restricted monitoring areas, and inefficiencies in observation [5], [6], [7].

Interferometric synthetic aperture radar (InSAR) technology represents a novel, all-weather earth observation method that can monitor ground deformation with large coverage, high resolution, and accuracy [8]. It has been widely used to mitigate the impact of various natural disasters, such as volcanoes [9], tsunamis [10], and earthquakes [11]. In addition, numerous scholars have employed InSAR technology to monitor subsidence in mining areas [12], [13]. However, due to the effects of decoherence, the quality of phase unwrapping varies across different regions, making rapid interpretation of the deformation map challenging [14], [15]. Although the use of novel time-series InSAR techniques like Multidimensional Small Baseline Subset (MSBAS) can delineate boundaries and 3-D deformations of one or multiple subsidence areas, there exist challenges related to prolonged data processing and low recognition efficiency [16], [17], [18]. With the advancement of technology, the interferometric width (IW) mode of contemporary synthetic aperture radar (SAR) satellites, like Sentinel-1A, can attain an image width of up to 250 km [19]. However, considering the stringent and extensive operational procedure of InSAR technology, detecting subsidence funnels with radii ranging from tens to hundreds of meters in a large-scale image is evidently time-consuming and labor-intensive. Therefore, there is a need to develop a new method for automatically detecting subsidence funnels using the InSAR interferogram on a large scale.

Recently, deep learning techniques, especially convolutional neural networks (CNNs), have experienced unprecedented and rapid development, notably in feature extraction and representation [20]. Deep learning provides innovative solutions for automatically identifying various geological disasters, encompassing volcanic and seismic hazard detection [21], [22], landslide hazard detection [23], urban ground subsidence

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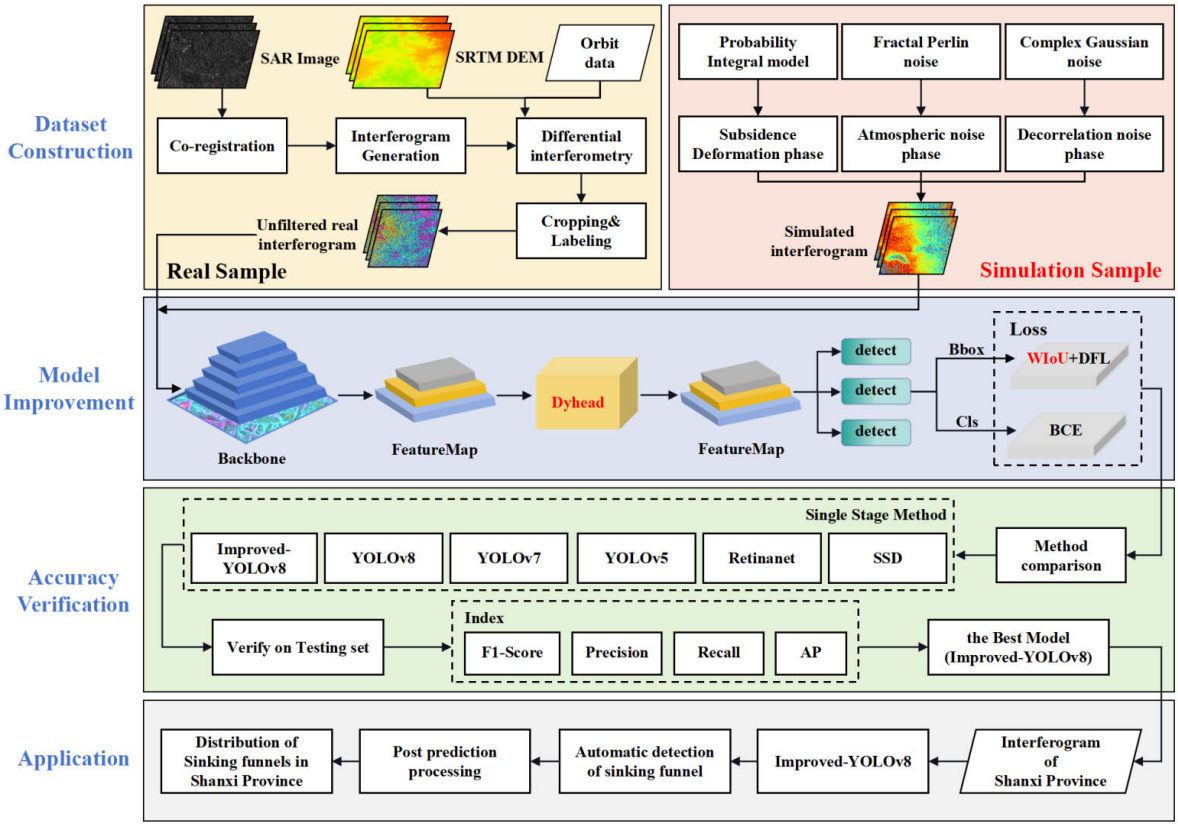


Fig. 1. Flowchart of research methods. The proposed dataset consists of two parts: the real sample set and the simulated sample set. For the training process, first, only the real sample set is used to train six models. Then, the simulated sample set is combined with the real sample set to retrain the models, and select the best model.

detection [24], and mining subsidence detection [25]. The target detection methods relying on CNNs have also drawn attention from scholars in the field of subsidence funnel detection in mining areas using InSAR interferograms. For example, Rotter et al. [26] engineered two alternative systems based on single shot multibox detector (SSD) and TinyYOLOv2 networks for the automatic detection of subsidence funnels induced by underground coal mining. Yu et al. [27] proposed a lightweight model for subsidence basin detection based on the YOLOv5 network, called Light You Only Look Once (YOLO)-Basin, significantly reducing model parameters while enhancing model accuracy. However, challenges persist in integrating deep learning with InSAR techniques. For instance, developing a subsidence funnel dataset with richer morphological features continues to be a challenge; how to further promote the detection accuracy of the deep learning model on InSAR interferograms; and assessing the generalization of trained deep learning models in practical applications, as well as their performance in ultralarge-scale detection tasks.

This article aims to propose a novel method for the automatic detection of subsidence funnels in mining areas on a large scale, utilizing deep learning algorithms and InSAR technology. Initially, a hybrid InSAR dataset is constructed by combining real interferogram data from mining areas with simulated interferogram data synthesized based on the probability integral method, Polin noise, and complex Gaussian white noise. Subsequently, on the basis of the YOLOv8 algorithm, a new subsidence funnel detection model (Improved-YOLOv8) is proposed by incorporating the attention-based adaptive detection head, dynamic head (Dyhead), and a more rational regression box loss function, Wise-intersection over union (WIoU). Finally, the effectiveness of the proposed method in this study is explored.

II. Methodology

This study proposes a research framework for the automatic detection of subsidence funnels over coal mining areas based on InSAR and deep learning. A subsidence funnel detection model was developed based on the improved YOLOv8 algorithm, which was jointly trained on real and simulated interferograms. The experimental methods in this article focus on three parts: the construction of a dataset containing real and simulated samples, the structural optimization based on the YOLOv8 model, and model accuracy verification. Fig. 1 shows the overall flowchart of this study.

A. Dataset Construction

Deep learning, characterized by its data-driven approach, encounters challenges in acquiring a sufficient quantity of real samples that meet specific requirements. This limitation results in heightened costs associated with providing the requisite target features to the network when relying exclusively on real sample sets for training [28], [29], [30]. The performance of models based on deep learning is largely dependent on

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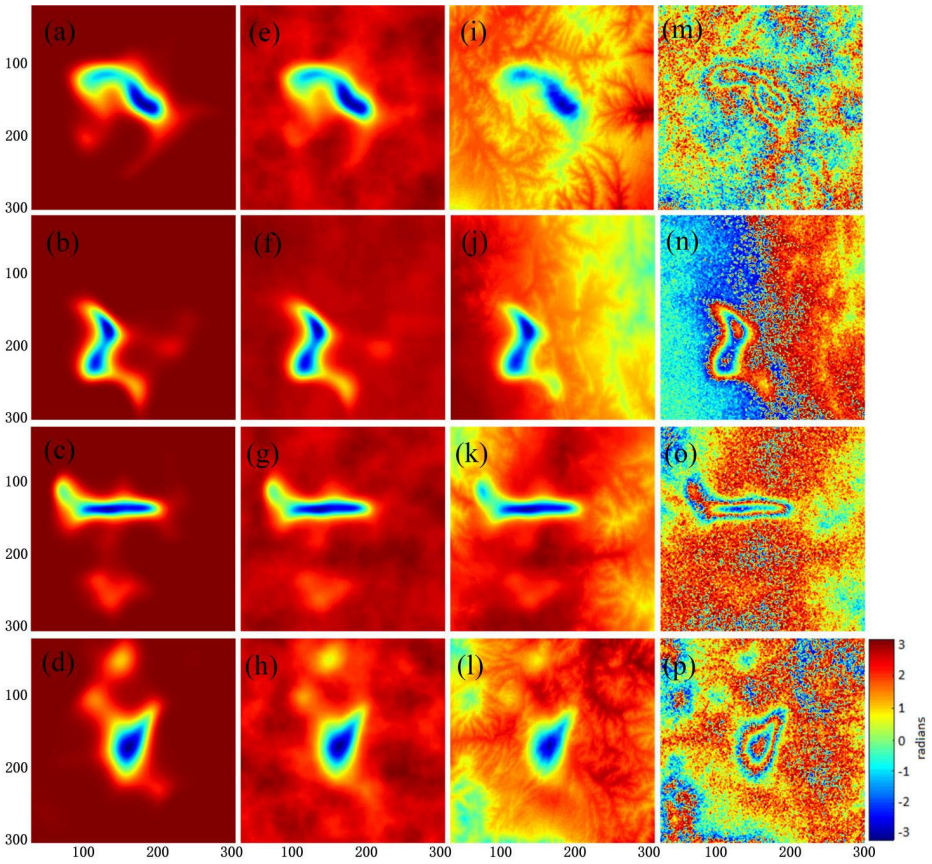


Fig. 2. Simulated interferogram samples. (a)–(d) Four examples of simulated subsidence funnel deformation phases with different patterns. (e)–(h) Integration of deformation phases and simulated atmospheric noise phases. (i)–(l) Integration of deformation phases, simulated atmospheric phases, and topographic phases. (m)–(p) Simulated deformation phases, simulated atmospheric phases, topographic phases, and simulated decorrelation noise phases, which were wrapped to obtain the final synthetic interferograms.

the features of training samples. In this study, a simulated interferogram sample set was generated and combined with a real interferogram sample set to expand the InSAR dataset, providing deformation samples with more diverse morphological features and ultimately improving detection accuracy.

1) Real Samples: Due to the lack of publicly available InSAR datasets with labeled subsidence funnels, a training and testing dataset needs to be constructed. Sentinel-1A SAR data from the Liulin mining area in Shanxi from January 6, 2018 to April 25, 2020 and the Yangquan mining area in Shanxi from January 3, 2019 to November 5, 2020 were selected as data sources. First, the interferometric processing step is performed, using differential radar interferometry (D-InSAR) to obtain 117 interferograms with a temporal baseline ranging from 12 to 84 days. When the input image of the deep learning network is too large, it will increase the burden on the computer hardware, and the target detection network has strict requirements on the size and scale of the input image. Therefore, the large-scale interferogram is segmented into 300 × 300 pixel-sized interferogram patches. Several interferogram patches obtained are screened to remove pure background patches that do not contain subsidence funnels. The subsidence funnel feature on the InSAR interferogram is relatively obvious, usually consisting of a series of approximately circular or elliptical interferometric fringes [31]. Using LabelImg software, the screened interferogram sub-images are manually labeled with mining subsidence funnels, generating image annotation files with bounding box coordinates. Finally, a real dataset consisting of 1489 real interferogram patches of 300 × 300-pixel size is obtained.

2) Simulated Samples: In order to enhance the fidelity of simulated interferograms to authentic data, this study incorporates three components into the phase of simulated interferograms: the mining subsidence deformation phase (D) simulated based on the probability integral model, the atmospheric noise phase (A) simulated based on fractal Perlin noise, and the decorrelation noise phase (C) simulated based on complex Gaussian white noise, as shown in Fig. 2.

After superimposing the three phases and wrapping them to $[-\pi, \pi]$, the simulated interferogram samples are obtained. The size of the simulated samples is set to 300 × 300 pixels, which is consistent with the real samples.

The probability integration method finds its foundation in stochastic media theory. Over decades of exploration by mining subsidence experts, this approach has evolved into a highly developed and extensively applied predictive technique in the analysis of rock movement. As elucidated by the fundamental principles of the probability integration method [32], [33], the subsidence deformation value (W) at any given point (x, y) on the surface induced by unit mining is expressed as

$$W(x, y) = \frac{1}{r^2} \cdot \exp\left[-\pi \frac{(x - x_i)^2 + (y - y_i + l_i)^2}{r^2}\right] \quad (1)$$

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$$r = \frac{H_0}{\tan\beta} \tag{2}$$

$$l_i = H_i \cdot \cot\theta \tag{3}$$

where r is the main influence radius, H_0 is the average mining depth, β is the main influence angle, θ is the maximum subsidence angle, (x_i, y_i) is the plane coordinate of the center point of the ith unit, and (x, y) is the coordinate of any point on the surface. By setting empirical parameters of the probability integral model, the surface deformation caused by mining can be simulated. Considering the uncertainty of the signal in the actual deformation result, the elastic transformation algorithm [34] is introduced to randomize the shape, position, and stripe density of the deformation. Elastic transformation is a random standard deviation (STD) of $(-1, 1)$ interval generated in each dimension of the pixel point, and the deviation matrix of each dimension is filtered with Gaussian filtering (0, sigma). Finally, the amplification factor alpha is used to control the deviation range. Therefore, $A(x, y)$ leads to $A'(x+\Delta x, y+\Delta y)$. Generally speaking, the smaller the alpha value, the larger the sigma value, the smaller the deviation, and the closer to the original image.

Fractal Perlin noise, created by overlaying Perlin noise with diverse frequencies and amplitudes, is employed to generate localized random atmospheric turbulence phases. To emulate the decorrelation effect, Complex Gaussian white noise is introduced. The noise generation process is delineated as follows:

$$n = \Omega \odot F(g) \tag{4}$$

where Ω denotes a random matrix conforming to a standard normal distribution and shares the dimensions of the training sample. The variable g represents a normalized phase gradient, subject to random modulation through the subsequent linear mapping function

$$F(g) = (\mathrm{NL} - \eta) \cdot g + \eta, \quad (0 \le \eta \le \mathrm{NL}). \tag{5}$$

Within this expression, parameter NL controls the STD of the maximum noise, while parameter η constrains the minimum NL and is randomly set during the simulation process. An independent noise matrix corresponding to the same NLs is generated for the real and imaginary components. Subsequently, these matrices are multiplied to yield a complex noise matrix, emulating the simulated interferogram [29].

By adjusting relevant parameters, a large number of simulated interferograms containing various shapes of subsidence funnels can be generated quickly. In the end, a total of 1000 simulated interferometric phase images with a size of 300 × 300 pixels were generated in this study to form a simulated sample set. However, synthetic data cannot fully simulate the fringe density and spatial distribution of noise and deformation signals in real interferograms.

Therefore, this study utilized 750 real interferogram samples and 750 synthetic interferogram samples, combined them to form a mixed InSAR dataset containing 1500 images, and retrained the model based on the hybrid dataset. About 3000 subsidence funnels are in the mixed dataset. Randomly, 85% of the images were selected as training samples, with the remaining 15% reserved for testing. Within the training samples, a subdivision into a training set and a validation set was executed at a ratio of 0.85:0.15. To mitigate overfitting resulting from limited sample datasets, images underwent rotational transformations at 15° intervals spanning from 0° to 90°. Additionally, both the original images and their rotated counterparts were subjected to horizontal and vertical flips to facilitate data augmentation.

B. Improved-YOLOv8 Model

YOLOv8, the latest in the YOLO series of target detection algorithms, has enhanced various components compared to its predecessors. In YOLOv8, anchor implementation is adopted with reference to YOLOX [35], providing advantages in handling irregularly shaped objects with elongated dimensions. When calculating the loss for positive sample matching, an adaptive multipositive sample matching method is adopted.

Due to a multitude of internal and external factors, the size of the subsidence funnel in the mining area commonly exhibits variation, even within identical regions. This makes the size of the detection target region in the interferogram different. However, the size of the detection box generated by the target detection network relies on the dimensions of the preset box, which is inflexible. Moreover, upon scrutinizing the subsidence funnel samples in the dataset, it is evident that the majority of samples exhibit distinct features and are easily discernible, categorized as simple samples; whereas a minority possess ambiguous features, increasing the likelihood of misclassification, labeled as difficult samples. This leads to a significant imbalance between easy and difficult samples. However, it is crucial for the model to learn and discern difficult samples to enhance the network's detection performance.

Building upon this, the study presents an improved algorithm based on YOLOv8 aimed at automatically detecting subsidence funnels within a large-scale interferogram. Initially, a Dyhead [36] module based on an attention mechanism is incorporated in the head section to accomplish adaptable detection of subsidence funnel targets spanning various scales. Subsequently, in the intersection over union (IoU) loss section, WIoU [37] substitutes the complete intersection over union (CIoU) present in the YOLOv8 model to address the imbalance between easy and difficult samples. Fig. 3 shows the main structure of the Improved-YOLOv8 algorithm.

1) Dyhead: Attention mechanism is a processing technique that directs focus toward pertinent information while filtering out extraneous, unrelated data. It can grasp the relative significance of diverse semantic levels and selectively amplify features according to the dimensions of an individual object. The Dyhead module unifies target detection head and attention mechanism. This approach cohesively integrates multiple self-attention mechanisms across feature levels, spatial locations, and output channels, enabling simultaneous attainment of scale perception, spatial perception, and task perception within a unified target detection head. This significantly enhances the representational capacity of the target detection head without inflating computational overhead. As shown in Fig. 4, the input to the detection head is a scaled feature pyramid F, represented by a 3-D tensor $F \in R^{L\times S\times C}$. Where L represents the number of layers in the pyramid,

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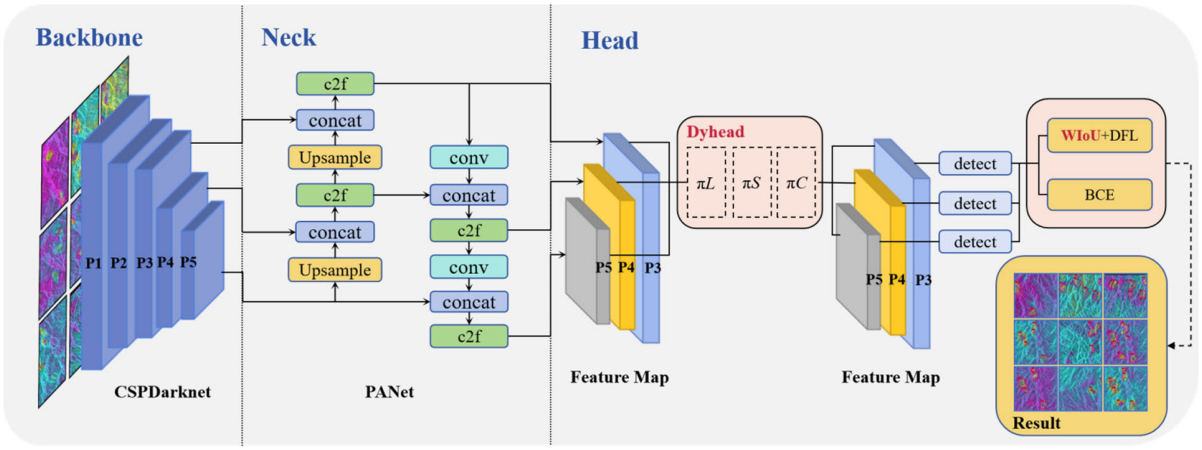


Fig. 3. Structure diagram of Improved-YOLOv8. P1, P2, P3, P4, and P5 are feature maps of different sizes. CSPDarknet is the backbone network, and PANet is the feature enhancement network.

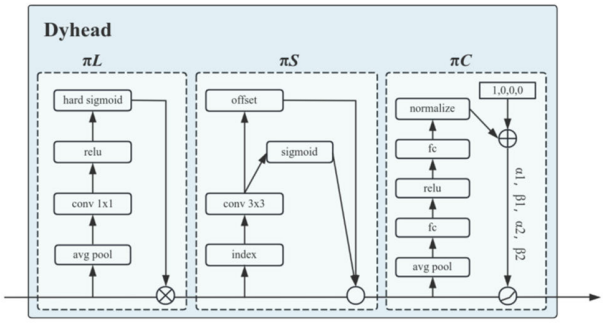


Fig. 4. Schematic of Dyhead structure. The three different process boxes πL, πS, and πC represent the scale-aware attention module, spatial-aware attention module, and task-aware attention module, respectively. $\alpha 1$, $\beta 1$, $\alpha 2$, and $\beta 2$ are hyperparameters of the activation function within the task-aware attention module.

$S = H \times W$, and H, W, and C represent the height, width, and channel number of the median horizontal feature, respectively. The scale-aware attention module operates along the horizontal dimension, dynamically amalgamating diverse scales, thereby adaptively incorporating the significance of varying feature levels; the spatial-aware attention module operates in the spatial dimension (i.e., height $\times$ width), adaptively pooling features across layers at identical spatial positions to acquire more discriminative representations; and the task-aware attention module operates within the channel, capable of dynamically activating or deactivating functional channels to flexibly facilitate diverse tasks [36].

2) WIoU: The loss function employed in YOLOv8 comprises three components, bifurcated into two branches: classification and regression. For classification, binary cross-entropy (BCE) loss is utilized, whereas CIoU bounding box loss and distribution focus loss (DFL) are adopted for regression. By assigning weights to these three loss functions, the total loss function is derived, expressed as

$$\text{Loss}^{\text{total}} = \alpha \text{Loss}^{\text{BCE}} + \beta \text{Loss}^{\text{DFL}} + \gamma \text{Loss}^{\text{CIoU}} \tag{6}$$

where α, β, and γ are weight coefficients.

The CIoU employed in YOLOv8 accounts for the bounding box regression's overlapping area, center point distance, and aspect ratio. However, its description of the aspect ratio as a relative value lacks clarity, neglecting the balance between challenging and straightforward samples. WIoU introduces a dynamically nonmonotonic focusing mechanism, utilizing "outlierness" instead of IoU for anchor box quality assessment, and provides a judicious gradient allocation strategy. Consequently, WIoU can concentrate on ordinary-quality anchor boxes, effectively addressing the imbalance between difficult and easy samples, and improving the overall performance of the detector [37].

WIoU is further divided into three versions: v1, v2, and v3, with specific computation formulas as follows:

$$L_{\text{WIoUv1}} = R_{\text{WIoU}} L_{\text{IoU}}$$
$$R_{\text{WIoU}} = \exp\left(\frac{(x - x_{\text{gt}})^2 + (y - y_{\text{gt}})^2}{(W_g^2 + H_g^2)^*}\right) \tag{7}$$

$$L_{\text{WIoUv2}} = r L_{\text{WIoUv1}}, \quad r = \left(\frac{L_{\text{IoU}}^*}{\overline{L}_{\text{IoU}}}\right)^{\gamma}, \ \gamma > 0 \tag{8}$$

$$L_{\text{WIoUv3}} = r L_{\text{WIoUv1}}, \quad r = \frac{\beta}{\delta \alpha^{\beta - \delta}}$$
$$\beta = \frac{L_{\text{IoU}}^*}{\overline{L}_{\text{IoU}}} \in (0, +\infty) \tag{9}$$

where (x, y) and $(x_{\text{gt}}, y_{\text{gt}})$ denote the center coordinates of the anchor box and the ground-truth box, respectively. W_g and H_g represent the width and height of the minimum bounding box. The superscript * indicates the detachment of W_g and H_g from the computation graph. β stands for outlierness, controlled by two hyperparameters α and δ, while r represents the gradient gain [37].

C. Baseline Methods

The most advanced deep learning target detection methods can be classified into two categories: region proposal-based methods and regression-based methods. Region proposal-based methods, also known as two-stage detection methods, divide the detection framework into two stages: candidate region extraction using algorithms and secondary refinement of the candidate bounding boxes. Therefore, they have higher detection accuracy. Representative methods include

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region-based convolutional neural network (R-CNN) [38], FastR-CNN [39], and Faster-RCNN [40]. Regression-based methods, also known as single-stage detection methods, do not require candidate region recommendation. They can determine the target category and locate the target in one stage, thus achieving high inference speed. Representative methods include the YOLO series [41] and RetinaNet. The aim of this study is to achieve rapid and automatic detection of subsidence funnels on large-scale interferograms. Although two-stage detection methods have high accuracy, they have complex model training and high computational requirements, making it difficult to achieve real-time detection. Therefore, five representative single-stage target detection methods were selected for this study: SSD [42], RetinaNet [43], YOLOv5 [44], YOLOv7 [45], and YOLOv8 [46] to improve experimental efficiency.

The SSD algorithm, proposed in 2016 and formally known as Single Shot MultiBox Detector, is a primary detection framework. SSD is inspired by the VGG-16 network model and its core lies in predicting convolutional feature maps of multiple scales to detect targets of varying sizes. The RetinaNet network, introduced in 2017, incorporates feature pyramid network (FPN) for feature map fusion and utilizes Focal loss to balance positive and negative samples. The YOLO series of target detection methods, originally proposed in 2016 and acronymized as YOLO, maintains a generally consistent prediction approach despite continual algorithm updates. This approach is typically divided into three components: Backbone, FPN, and Yolo Head. Furthermore, due to the diversity of object types and shapes in natural images, heavy networks are often used as the backbone. However, the shape of the subsidence funnel is relatively simple, and the detection category is single, so a heavy network is not required for subsidence funnel detection [27]. Therefore, in this study, lightweight versions of YOLO series models were uniformly selected. Specifically, YOLOv5 and YOLOv8 were selected with the "*s*" version model, while YOLOv7 was selected with the "*l*" version model.

All experiments were trained and tested on a computer equipped with an AMD Ryzen7-5800H CPU (3.20 GHz) and an NVIDIA GeForce RTX 3060 graphics card. PyTorch (GPU version) was used as the experimental deep learning framework, with CUDA 11.6 and CuDNN 8.3 for GPU parallel computing and GPU acceleration. The Adam optimizer was used for network parameter updates and optimization. The initial learning rate for training was set to 1e-4. The dynamic adjustment method for the learning rate was set to the cosine annealing algorithm. The mini-batch size was set to 9 and the training iteration number was set to 100.

D. Accuracy Verification

The main metric used to evaluate the performance of deep learning models is mean average precision (mAP). This metric finds application in many other object detection studies to illustrate the accuracy and recall trends of various training models [47]. The mAP is essentially the average of the AP values for each class, and the average precision (AP) is calculated using the difference AP evaluation method, which represents the area under the precision–recall curve. The formula is

$$\mathrm{AP} = \left(\frac{1}{n} \sum_{\left(r \epsilon \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}, 1\right)} P_{(r)} \right) \tag{10}$$

$$\mathrm{mAP} = \frac{1}{N} \sum_{k=1}^{k=N} \mathrm{AP}_k \tag{11}$$

where n represents the number of detection points, $P_{(r)}$ represents the value of accuracy at a recall rate of r; N is the number of categories, and AP_k is the average precision of category k. Because the target category detected in this study is only the "subsidence funnel," mAP is actually equal to AP in this study, so AP is used as the main accuracy verification indicator. AP_{50} and AP_{75} represent IoU thresholds of 0.5 and 0.75, respectively. AP_{50-95} represents the average AP value under different IoU thresholds with a step size of 0.05 and an IoU threshold range of 0.5–0.95. In addition, auxiliary evaluation metrics such as precision, recall, and $F1$-score are represented mathematically as follows:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{12}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{13}$$

$$F1 = 2\frac{\mathrm{Recall} * \mathrm{Precision}}{\mathrm{Recall} + \mathrm{Precision}} \tag{14}$$

where TP represents true positives, referring to the situation where an object is correctly predicted to be a positive sample. FP represents false positives, which refers to detecting an invalid object as a positive sample. FN represents false negatives, which means that a valid object is detected as a negative sample.

III. Results and Analysis

A. Study Area

Shanxi is a major coal mining province in China, with coal mines widely distributed across various cities. The province is characterized by numerous basins, with a majority of the subsidence funnels concentrated at the basin's periphery, while only sporadic occurrences are observed in the basin's center. According to analysis, the peripheral basins have thinner mineral resource cover layers, making mining activities more intensive, resulting in an increased number of subsidence funnels. In contrast, the basin's center has a thicker cover layer, coupled with advanced deep mining techniques, leading to relatively weaker surface settlements [48]. The real interferograms used for training in this study were obtained from the Yangquan and Liulin coal mines in Shanxi. The Liulin coal mine is located in the central-western part of Shanxi, as shown in Fig. 5(b). The Yangquan coal mine is located in the central-eastern part of Shanxi, as shown in Fig. 5(c). Both mines have numerous coal mines and are rich in coal resources, with proved coal reserves exceeding 10 billion tons [49]. Some local mine interferograms that include subsidence funnels are shown in Fig. 5(d) and (e).

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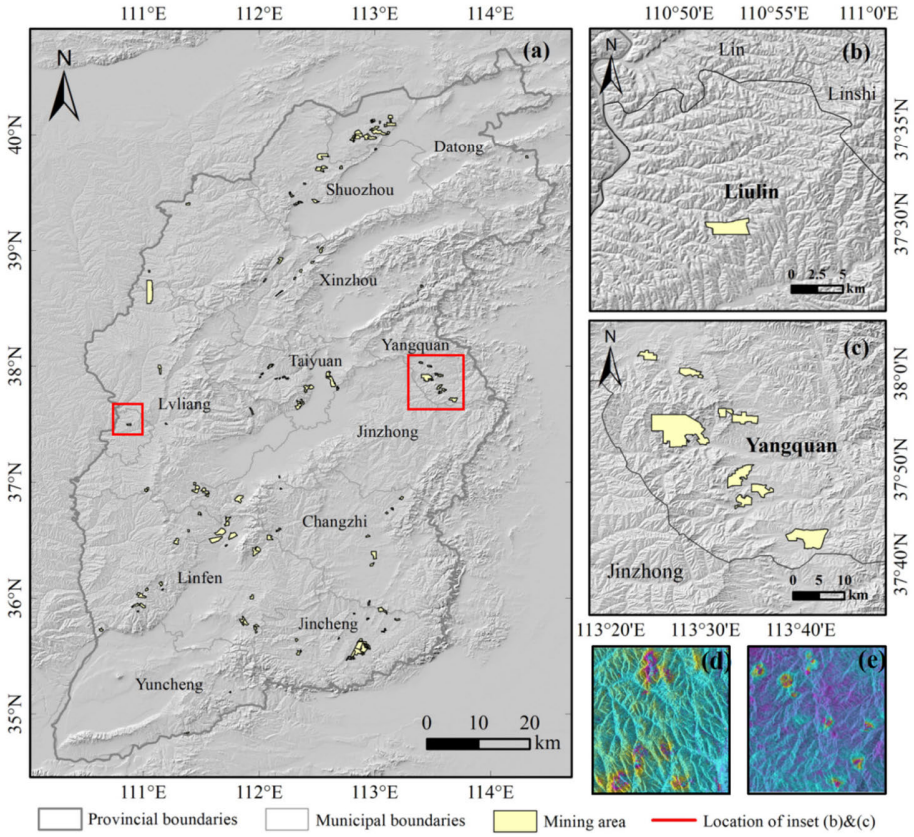


Fig. 5. (a) Location map of the main mining areas in Shanxi. (b) and (c) Schematics of the Liulin and Yangquan mining areas, respectively. (d) and (e) Interferograms of some subsidence funnels in the Liulin and Yangquan mining areas, respectively.

TABLE I
COMPARISON OF DETECTION ACCURACY OF FIVE MODELS TRAINED ON DIFFERENT DATASETS

Method	Data Set	AP_{50}	AP_{75}	AP_{50-95}
YOLOv5	real	0.889	0.529	0.496
	real+simulate	**0.900**	**0.557**	**0.512**
YOLOv7	real	0.893	0.529	0.495
	real+simulate	**0.906**	**0.589**	**0.528**
YOLOv8	real	0.898	0.572	0.521
	real+simulate	**0.908**	**0.589**	**0.532**
SSD	real	0.892	0.480	0.466
	real+simulate	**0.903**	**0.469**	**0.473**
RetinaNet	real	0.759	0.388	0.391
	real+simulate	**0.783**	**0.407**	**0.408**

B. Optimizing Dataset Comparison

To verify the improved detection accuracy of the mixed InSAR dataset proposed in this study for model subsidence funnels, five target detection models were retrained using a dataset constructed from a combination of simulated interferogram samples and real interferogram samples, and compared with models trained using only real sample sets. The comparison results of the detection accuracy of the training models with the same number of two different datasets are shown in Table I.

The experimental results show that after retraining the model using the proposed hybrid dataset, the detection accuracy of five object detection models for detecting sink funnels has been improved compared to the model trained using only the real dataset. The AP-related indicators of YOLOv5, YOLOv7, YOLOv8, and SSD have all increased by about 1%, while the AP_{50}, AP_{75}, and AP_{50-95} of RetinaNet increased by 2.4%, 1.9%, and 1.7%, respectively. As can be seen from Table I, among the five basic models, YOLOv8 method is the most accurate method for detection. Taking the YOLOv8 model as an example, Fig. 6 demonstrates the detection effect of the subsidence funnel in the same area.

C. Comparison of Deep Learning Methods

This article proposed an enhanced model derived from YOLOv8. By introducing an adaptive detection head Dyhead based on attention mechanism and a more rational regression box loss function WIoU, the proposed model aims to address the concerns of multiscale targets and imbalanced samples in detecting subsidence funnels in interferograms. In order to solve the imbalance problem between difficult and easy samples and improve detection accuracy, WIoU is introduced to replace CIoU in the regression box loss function of the YOLOv8 model. To ascertain the distinctive adaptability of WIoU within the scope of this investigation, a comparative analysis is conducted with four other regression box loss functions, including CIoU, which are integrated into the model for

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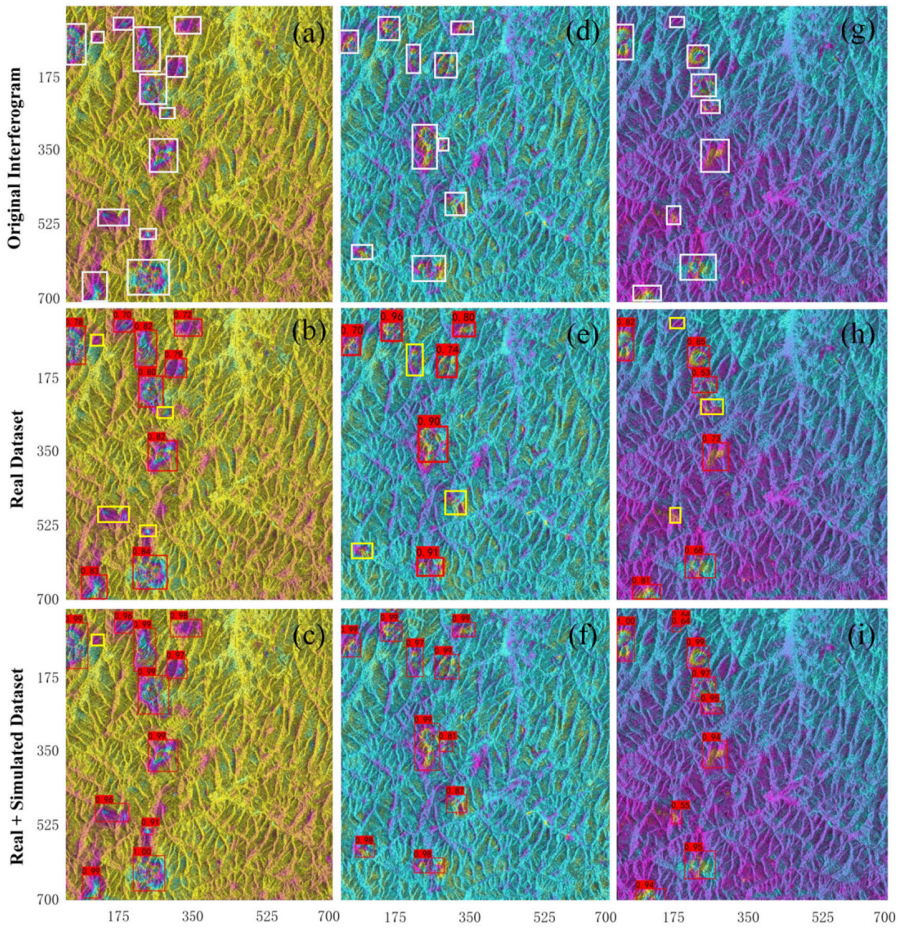


Fig. 6. Detection results of YOLOv8 trained on real and hybrid datasets. (a), (d), and (g) Three original interferograms to be detected, with white rectangles indicating ground truth. (b), (e), and (h) Detection results of YOLOv8 trained only on real datasets, with red rectangles indicating detection results and yellow rectangles representing missed targets. (c), (f), and (i) Detection results of YOLOV8 trained on hybrid datasets.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT REGRESSION BOX LOSS MODULES

Method	Data Set	Recall	Precision	F1 Score	AP_{50}(%)	$AP_{50\text{-}95}$(%)
YOLOv8(CIoU)	real	76.02%	90.32%	0.83	89.8	52.1
WIoU	real	**77.12%**	91.62%	**0.84**	**90.4**	**52.8**
FocalEIoU	real	67.71%	**93.71%**	0.79	89.4	52.2
SIoU	real	76.49%	91.73%	0.83	90.2	52.4
MPDIoU	real	73.98%	91.47%	0.82	89.8	52.0

experimental validation. The outcomes of these experiments are presented in Table II.

The experimental results show that WIoU outperforms other advanced regression box loss calculation methods such as CIoU, FocalEIoU, SCYLLA-IoU (SIoU) and minimum point distance intersection over union (MPDIoU) which are included with YOLOv8. The AP_{50}, AP_{50-95}, and $F1$-score of the YOLOv8 model using WIoU as the regression box loss function can reach 90.4%, 52.8%, and 0.84, respectively, all of which are superior to the other four methods. Although WIoU is lower in precision than FocalEIoU and SIoU, its recall is higher than the latter two. In the task of detecting sedimentation funnels, missed detections are more unacceptable than incorrect detections, indicating that the improvement in Recall is more crucial to this study. In summary, WIoU was selected. Fig. 7 displays the variations in loss and accuracy throughout the epochs in the model training phase.

To assess the detection accuracy of the proposed Improved-YOLOv8 model in identifying subsidence funnels, a comparative analysis was conducted against five object detection methods, including the classic YOLOv8 model. The findings are presented in Tables III and IV.

Based on the experimental results, it can be seen that the Improved-YOLOv8 model proposed in this article outperforms other comparative models in terms of AP metrics when trained on both real and hybrid datasets, and the model has smaller Params and FLOPs metrics, and the highest FPS metric, indicating that it has a smaller model complexity and the

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TABLE III
COMPARISON OF DETECTION ACCURACY OF TEST SETS TRAINED ON REAL DATASETS. BOLD NUMBERS INDICATE THE BEST VALUES IN THE COLUMN

Method	Data Set	AP_{50} (%)	AP_{75} (%)	AP_{50-95} (%)	size	Params(M)	FLOPs(G)	FPS
Improved-YOLOv8	real	**90.8**	**59.9**	**53.0**	416x416	10.576	12.634	**93.9**
YOLOv5	real	88.9	52.9	49.6	416x416	**7.064**	**6.961**	66.8
YOLOv7	real	89.3	52.9	49.5	416x416	37.195	44.41	57.9
YOLOv8	real	89.8	57.2	52.1	416x416	11.136	12.103	91.2
SSD	real	89.2	48.0	46.6	300x300	23.612	60.756	81.3
RetinaNet	real	75.9	38.8	39.1	300x300	36.330	37.281	45.4

TABLE IV
COMPARISON OF DETECTION ACCURACY OF TEST SETS TRAINED ON HYBRID DATASETS

Method	Data Set	AP_{50} (%)	AP_{75} (%)	AP_{50-95} (%)
SSD	hybrid	90.3	46.9	47.3
RetinaNet	hybrid	78.3	40.7	40.8
YOLOv5	hybrid	90.0	55.7	51.2
YOLOv7	hybrid	90.6	58.9	52.8
YOLOv8	hybrid	90.8	58.9	53.2
Improved-YOLOv8	hybrid	**92.0**	**60.0**	**54.1**

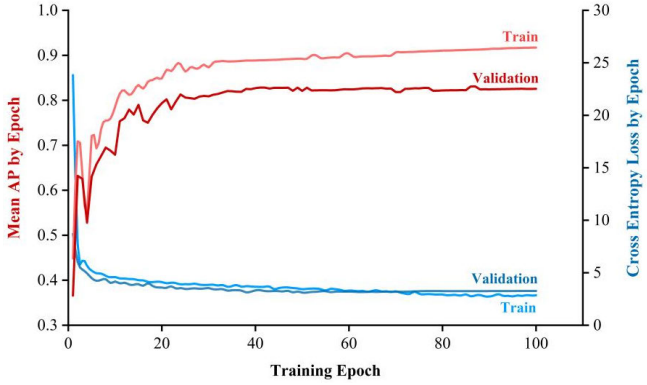


Fig. 7. Loss and accuracy in training and validation by epoch. The descent of the model's loss and its convergence process.

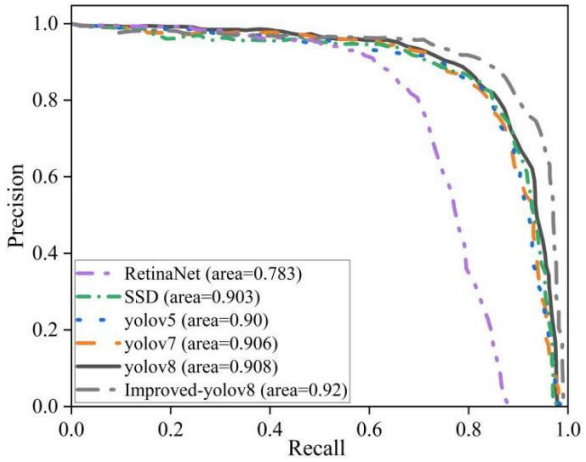


Fig. 8. Comparison of AP_{50} on the mixed test set of improved model and other methods. Compared with other tested models, the Improved-YOLOv8 model shows significantly better performance.

fastest model transmission speed. Overall, under the same experimental method, the Improved-YOLOv8 model is the best method among the six deep learning methods.

The Improved-YOLOv8 model achieves AP_{50}, AP_{75}, and AP_{50-95} of 92.0%, 60.0%, and 54.1%, respectively when trained on hybrid datasets. In addition, combining the results of Tables III and IV, it can be seen that the Improved-YOLOv8 model trained on the hybrid dataset achieves the highest detection accuracy among all methods in this study. Compared with the YOLOv8 base model trained on the real dataset, the AP_{50}, AP_{75}, and AP_{50-95} for detecting the subsidence funnel are increased by 2.2%, 2.8%, and 2.0%, respectively. Fig. 8 shows the detection effect comparison of the Improved-YOLOv8 and other models on the same local area of the subsidence funnel when trained on hybrid datasets.

According to the qualitative comparison results of the six deep learning methods shown in Fig. 9, it can be seen that in the same region, the Improved-YOLOv8 model can detect the subsidence funnel that is ignored by other models. Despite some errors, Improved YOLOv8 demonstrated superior performance in accurately detecting the subsidence funnel in Fig. 9 compared to other models evaluated in this study.

IV. DISCUSSION

A. Parameters of Improved Model and Ablation Experiment

The performance enhancement of the Dyhead module is closely associated with the number of stacked layers [36]. Additionally, WIoU exists in three versions: v1, v2, and v3, with WIoUv3 requiring the determination of hyperparameters α and δ [37]. Therefore, to assess the impact of the quantity of stacked Dyhead modules and the versions of WIoU along with their hyperparameters on the model's performance enhancement, we conducted several comparative experiments.

The results of the two comparative experiments are shown in Fig. 10. From the experimental results, it can be seen that

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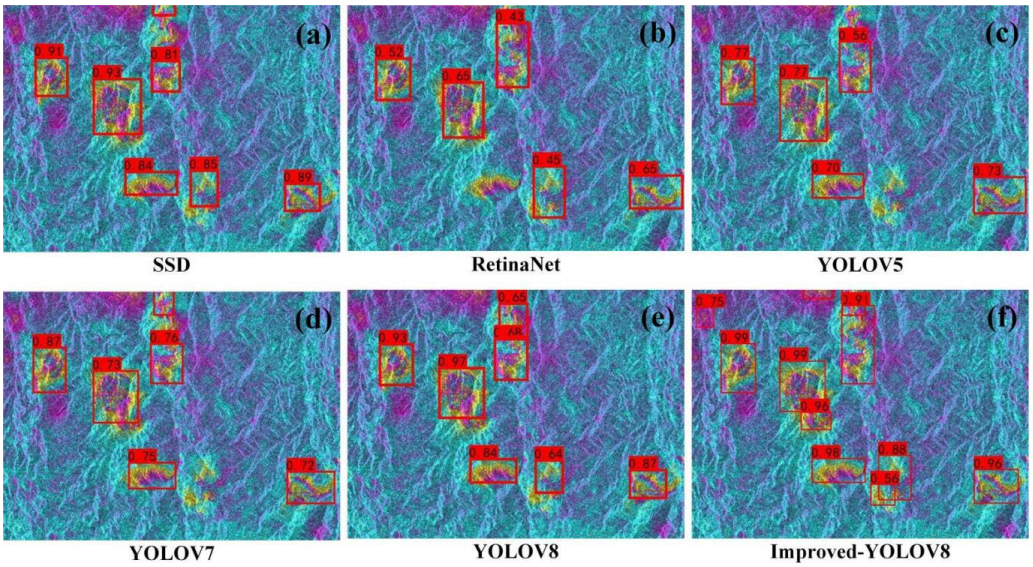


Fig. 9. Comparison of the detection performance of the improved model with other models for the detection of local subsidence funnels. (a)–(f) Detection results of the six methods and the red rectangular box in the figure represents the subsidence funnel detected from the InSAR interferogram. The number above the rectangular box represents the probability that the detected area belongs to the subsidence funnel.

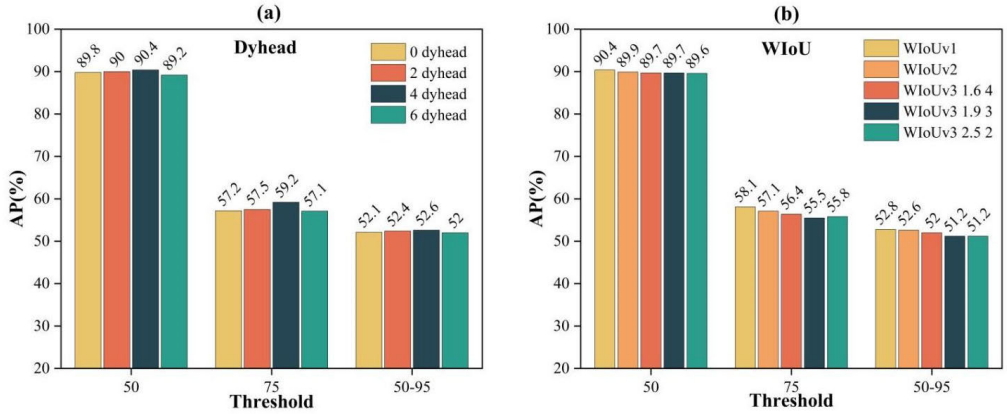


Fig. 10. Comparison results of the impact of different parameters on the accuracy of the model. (a) AP comparison of the model when different numbers of Dyhead modules are stacked. (b) AP comparison of the model when different versions of WIoU and different parameters of WIoUv3 are used.

in this study, when the number of Dyhead modules stacked on YOLOv8 reaches 4, the model achieves the highest detection accuracy for the subsidence funnel, with AP_{50}, AP_{75}, and AP_{50-95} reaching 90.4%, 59.2%, and 52.6%, respectively. Therefore, the Dyhead module introduced in this study refers to four stacked Dyhead modules. When using WIoUv1 as a regression box loss function instead of CIoU, the model achieves the highest detection accuracy for the subsidence funnel, with AP_{50}, AP_{75}, and AP_{50-95} reaching 90.4%, 58.1%, and 52.8%, respectively. Therefore, the WIoU module referred to in this study refers to WIoUv1.

Furthermore, to demonstrate the individual improvement effects of the two enhanced modules introduced in this article on the YOLOv8 model, and to further enhance the model's performance by combining them, we conducted ablation experiments on the Improved-YOLOv8 model. The results are shown in Table V.

Table V visually shows the changes in model accuracy after adding different modules. Based on the results of the ablation experiments, we can conclude that the introduction of both enhanced modules of Improved-YOLOv8 has significantly improved the model. When these two modules are combined, there is a more pronounced increase in model performance. As described in the Methodology Section, the Dyhead module aids the network in the adaptive detection of subsidence funnels of various scales from interferograms, while WIoU can to some extent address the issue of sample balance between easy and difficult examples.

B. Robustness and Generalizability of Improved-YOLOv8

Due to speckle effects and various other forms of noise, coherent speckle noise is commonly present in InSAR interferograms, exerting adverse effects on subsequent processing steps [50]. Numerous advanced speckle filters are typically employed to mitigate these noise sources [51], [52], [53], [54]. However, considering the practical scenarios where interferogram quality often remains suboptimal even after filtering, resulting in residual noise, both real and simulated samples

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TABLE V
ABLATION EXPERIMENT

Method	Data Set	Recall	Precision	F1	AP_{50} (%)	AP_{75} (%)	AP_{50-95} (%)
YOLOv8	real	76.02%	90.32%	0.83	89.8	57.2	52.1
YOLOv8+Dyhead	real	78.37%	91.58%	0.84	90.4	57.5	52.6
YOLOv8+WIoU	real	77.12%	91.62%	0.84	90.4	57.1	52.8
YOLOv8+Dyhead+WIoU	real	**79.15%**	90.83%	0.85	90.8	59.9	53.0
YOLOv8+Dyhead+WIoU	real+simulate	78.23%	**93.31%**	**0.85**	**92.0**	**60.0**	**54.1**

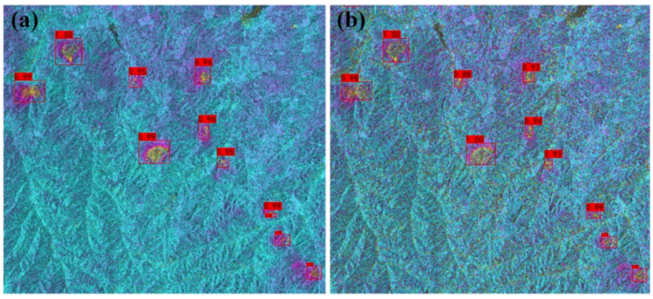


Fig. 11. Detection results of Improved-YOLOv8 model for filtered and unfiltered interferograms. (a) Interferogram after applying Goldstein filtering and (b) unfiltered interferogram. The red rectangular box is the model detection box.

utilized for training in this study were devoid of speckle filtering. As illustrated in Fig. 11, the trained Improved-YOLOv8 model was applied to detect subsidence funnels in the same local region of interferograms with and without speckle noise. The results demonstrate that the model trained on unfiltered interferograms in this study exhibits robustness against speckle noise.

In interferograms of mountainous regions, the presence of regular dense stripes, attributed to SAR image layover, is common. Conversely, in proximity to water areas, a significant amount of decorrelation noise is frequently observed. These distinctive characteristics pose challenges to the model's extraction of subsidence funnel features, leading to a notable decline in detection accuracy within these specific areas. To evaluate the generalization of the proposed model to different terrain types, local subsidence funnel detection was performed in mountainous regions, near water regions, and plain regions using the Improved-YOLOv8 model and four other comparative models, as shown in Fig. 12. The results show that the Improved-YOLOv8 model proposed in this study can achieve better detection of subsidence funnels in mountainous, near-water, and plain regions, and its generalization performance is superior to the other four compared models.

To evaluate the generalization ability of the proposed model on multisource data, we tested it using interferograms generated from RADARSAT-2 SAR data. We selected two representative large-scale interferograms of mining areas, as shown in Fig. 13(a) and (d). Both maps have a size of 3155 × 3813 pixels and a temporal baseline of 24 days.

Fig. 13(a) represents the period from December 31, 2015 to January 24, 2016, while Fig. 13(d) represents the period from January 24, 2016 to February 17, 2016. It is evident that, despite some typical errors (residual topographic and orbit phase error), the model accurately detected the majority of the subsidence funnels in the image. Consequently, it can be concluded that although Improved-YOLOv8 is specifically designed and applied for Sentinel-1 images, it has the potential to adapt to other sensors. However, to achieve optimal performance, retraining is required for interferograms with different resolutions, wavelengths, and application scenarios.

C. Sensitivity Analysis of Temporal Baseline

When conducting interferometric processing on SAR data, choosing an appropriate temporal baseline for the interferometric pair is essential. It is observed that with an increasing temporal baseline, the deformation of the same subsidence funnel target intensifies, leading to more pronounced features on the interferogram, as illustrated in Fig. 14. However, an excessively large temporal baseline would cause severe temporal decorrelation, obscuring the features of the subsidence funnel, as depicted in Fig. 14(l) and (p).

To explore the impact of the temporal baseline of the interferogram on the accuracy of model detection, we used sensitivity analysis [54] to expose the model to a more diverse set of data and statistically determined the approximate range of detectable settlement funnel deformation. By generating interferograms with different temporal baselines (from 12 to 48 days, in increments of 12 days) for the same local region, we tested the trained model for sensitivity analysis. The results are shown in Fig. 15.

It is evident that as the temporal baseline of interferograms extends from 12 days to 24 days, there is a noticeable increase in the model's detection accuracy. However, further extension beyond 24 days to 36 days results in nearly identical detection accuracy. Yet, when the temporal baseline is prolonged from 36 days to 48 days, there is a sharp decline in detection accuracy. Consequently, in the practical implementation of subsidence funnel detection, choosing either a 24-day or 36-day temporal baseline is recommended based on specific circumstances to attain optimal detection effectiveness.

In addition, the test set in this study has a temporal baseline ranging from 12 to 84 days. All sizes of subsidence funnels in the test set were detected, with the smallest detectable displacement of less than 2.8 cm and the largest detectable displacement of more than 8.4 cm. The detectable displacement range refers only to the approximate displacement estimated on the identified subsidence funnels.

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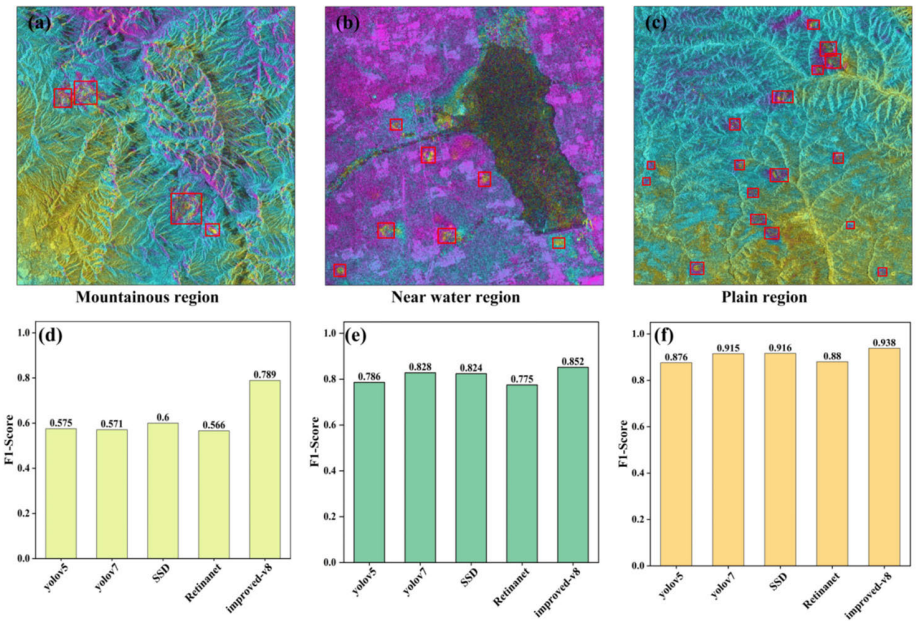


Fig. 12. First three images show the interferograms of three different regions. (a) Mountainous region. (b) Near water region. (c) Plain region. The subsidence funnels are indicated by red rectangle frames. The three bar graphs in the second row (d)–(f) demonstrate the detection accuracy of Improved-YOLOv8 and other models in corresponding areas for subsidence funnel detection.

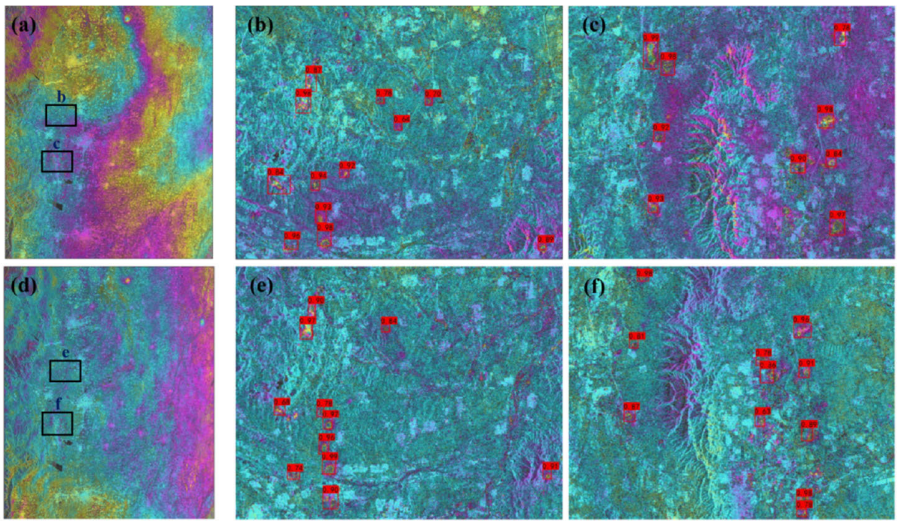


Fig. 13. RADARSAT2 data test results. (a) and (d) Two interferograms generated from RADARSAT2 SAR data. (b) and (c) Detection results of some subsidence funnels in interferogram (a). (e) and (f) Detection results of some subsidence funnels in interferogram (d).

D. Application of Improved-YOLOv8 in the Whole Shanxi

The ongoing expansion of mining operations would cause substantial damage in both the mining area and its adjacent environment [29]. Consequently, consistent monitoring of land subsidence across the province holds utmost significance within this region. Additionally, in an effort to assess the efficacy of the Improved-YOLOv8 model trained on a hybrid dataset within real-world contexts, we employed the proposed model to detect subsidence funnels within the mining zones of Shanxi, a major long-term mining province in China.

As illustrated in the blue rectangle in Fig. 16, a total of 16 scenes of SAR data from three consecutive orbits were selected to generate three large-scale interferograms, covering the lower, middle, and upper regions of Shanxi. To maintain coherence, the interferometric pair temporal baseline is standardized to 24 days. The SAR images were acquired between December 1, 2015, and January 1, 2016.

The center location of the subsidence funnel is determined based on the top-coordinate of the model detection frame, enabling the extraction of the subsidence funnel distribution across Shanxi, depicted in Fig. 16. In combination with the scope of key mining areas within Shanxi, the distribution pattern of subsidence funnels can be analyzed.

Fig. 16(A) shows the detection results of the model in the entire Shanxi. Using the trained Improved-YOLOv8, we detected 952 subsidence funnels in various parts of Shanxi. In addition, Fig. 16(B) shows the statistics of the number of subsidence funnels in 11 prefecture-level cities in Shanxi. The

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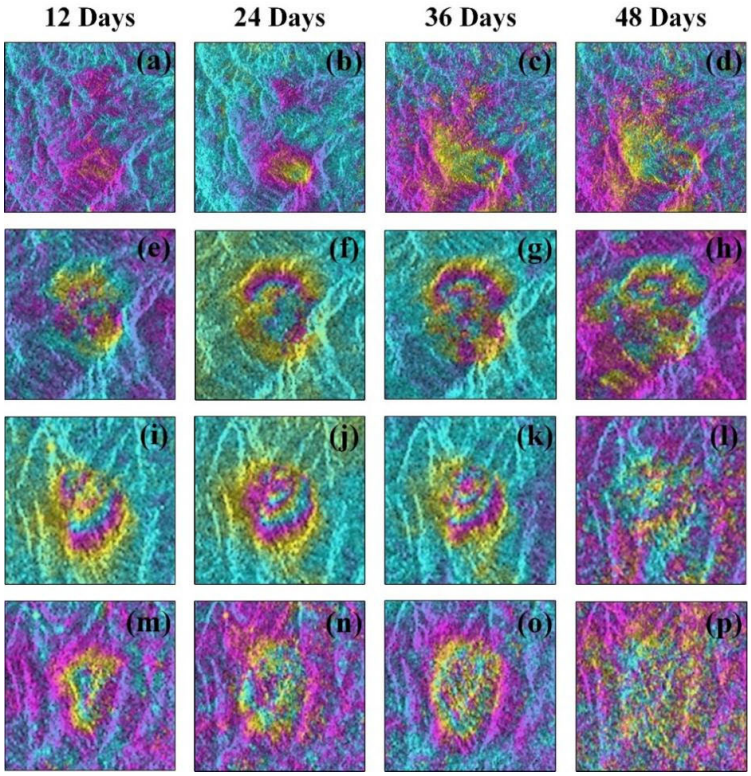


Fig. 14. Images from left to right in each row represent the characteristics of the same sedimentation funnel in the interferogram at temporal baselines of (a)–(d) 12 days, (e)–(h) 24 days, (i)–(l) 36 days, and (m)–(p) 48 days, respectively.

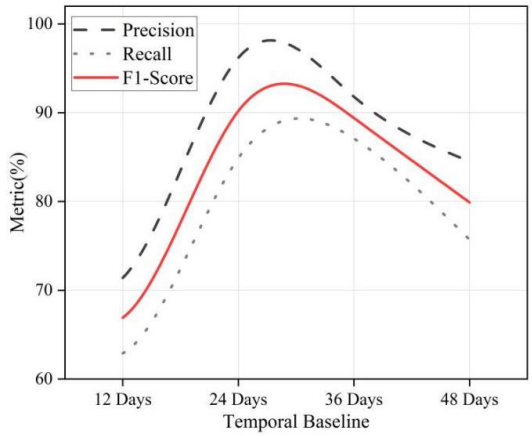


Fig. 15. Performance of the model over temporal baselines.

results show that Taiyuan, Linfen, and Lvliang are the top three cities with the most subsidence funnels in Shanxi, while Yuncheng has the least distribution of subsidence funnels. In 2022, Wu Zhipeng's team used DDNet to monitor the local collapse area in the central region of Shanxi from September 10, 2018 to November 16, 2019, and produced a subsidence area distribution map [29]. The detection results of this work in the central region of Shanxi are highly consistent with Wu's distribution map, proving the authenticity and credibility of our results. In addition, subsidence funnel detection was also performed in the upper and lower regions of Shanxi in this work, and a more complete subsidence funnel distribution diagram of Shanxi was obtained.

According to the distribution of key mining areas in Shanxi, the detected areas of intense subsidence funnels are largely consistent with the distribution of these key mining areas. As shown in Fig. 16(a) and (b), the subsidence funnels in and around the mining areas are widely distributed, further confirming the close association between mining activities and the formation of subsidence funnels. By observing the distribution of the subsidence funnels, can easily determine the general scope and progression direction of mining activities. Therefore, utilizing InSAR interferometric maps for rapid deformation area detection plays a crucial role in monitoring ongoing mining activities.

E. Limitations and Future Work

To facilitate comparison with other studies on mining-induced subsidence funnel detection based on InSAR and CNN, we have summarized and compiled Table VI to provide a clearer comparison in terms of dataset, network architecture, and accuracy. Specifically, our study has a certain degree of innovation in both the dataset and model. We combined real and simulated interferogram datasets, which enhances dataset diversity compared to other studies. Our Improved-YOLOv8 model achieved a 92.0% AP_{50} accuracy on Shanxi data, outperforming most single-stage models and rivaling some two-stage models. But there are also some limitations. Our datasets are mainly from Shanxi. Future work will include more regions to enhance generalization. Additionally, differences in evaluation criteria and datasets across studies pose challenges for direct comparison. Further validation on diverse datasets is needed in the future.

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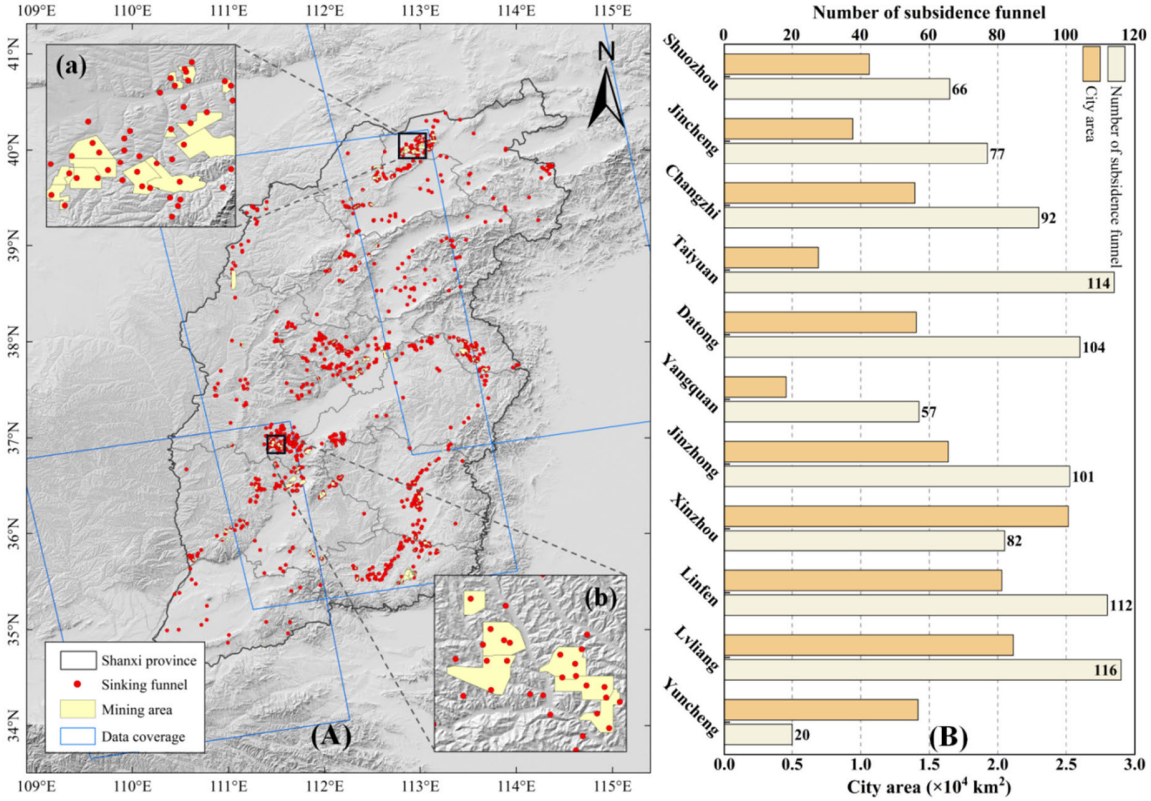


Fig. 16. Distribution of subsidence funnels in Shanxi Province from December 2015 to January 2016. (A) Location of subsidence funnels and key mining areas. The thumbnail (a) and (b) show two key mining areas and subsidence funnel distribution in Datong and Lvliang, respectively. (B) Histogram of subsidence funnels in 11 cities in Shanxi.

TABLE VI
RESEARCH ON MINING SUBSIDENCE FUNNEL DETECTION BASED ON INSAR AND CNN

Literature	Model	Single/Two Stage	Study Area	Dataset	Accuracy Rate
(P. Rotter, et al.,2021)	SSD & TinyYOLOv2	Single Stage	/	Real Interferogram	0.89(AUC) & 0.87(AUC)
(Y. Yu, et al.,2022)	Light YOLO-Basin (YOLOv5)	Single Stage	Jining and Huaibei Coalfields	Real Interferogram	90.64% (AP_{50})
(L. Wang, et al.,2022)	CNN-AFSA-SVM	/	Huainan Mining Area	Real Interferogram	88.30%
(Z. P. Wu, et al.,2022)	DDNet	/	Central Shanxi Province	Simulated Interferogram	99.30%
(J. H. Cai, et al.,2023)	Faster R-CNN	Two Stage	Southern Guizhou Province	Real Interferogram	94.00%
(K. L. He, et al.,2024)	Mask R-CNN	Two Stage	Southwestern Shanxi Province	Real+Simulated Interferogram	91.48%
Our Study	**Improved-YOLOv8**	**Single Stage**	**Shanxi Province**	**Real+Simulated Interferogram**	**92.0% (AP_{50})**

In order to provide more diverse settlement funnel shape features to the model during training, this article simulates a large number of synthetic samples based on the probability integral method, Polin noise, and complex Gaussian white noise, and combines them with real samples to obtain a mixed InSAR dataset. The Improved-YOLOV8 model has better generalization performance compared to other comparative models for regions with different land surface types. However, due to the effects of layover and large amounts of decorrelation noise [55], the detection accuracy of the model inevitably decreases in mountainous and near-water regions. This is due to the fact that the characteristics of layover and completely decorrelation areas such as water areas in the interferogram are not considered in the simulation, so the signals caused by layover and water areas can confuse the model [54]. In future work, the dataset will undergo fine-tuning operations to further enhance the model's generalizability. The plan involves adjusting the parameters of complex Gaussian noise in random regions within the simulated samples to simulate the complete decorrelation effect of water areas. Additionally,

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based on a scale model derived from natural scenes, the interferometric phase will be obtained using a multichannel array SAR simulation method [56].

In addition, the Improved-YOLOv8 model proposed in this study mainly optimizes the final detection process of the model by making changes to the head part of the model, thereby improving the performance of the model. However, because the model does not make improvements in the backbone and neck parts, it does not further optimize the feature extraction and enhancement process of the model. During the model testing process, we found that some factors that affect the performance of the model are fundamentally related to feature extraction [54]. Therefore, in the future, we will focus on promoting the model's multiscale feature extraction capabilities. For example, by introducing a Dilation-wise Residual (DWR) attention module [57], we can enhance the multiscale receptive field features, facilitate small target detection, and further improve the detection accuracy of the model.

V. Conclusion

In this article, we proposed a novel method combining deep learning object detection algorithms with InSAR methods to automatically detect subsidence funnels in mining areas within large-scale regions. The proposed Improved-YOLOv8 algorithm, incorporating Dyhead and WIoU, addresses multiscale target challenges and sample imbalance in interferograms. Trained on real data, the Improved-YOLOv8 model achieved 90.8%, 59.9%, and 53.0% in AP_{50}, AP_{75}, and AP_{50-95}, respectively, surpassing the baseline YOLOv8 model.

Using a hybrid dataset comprising simulated and real interferometric patterns significantly enhanced subsidence funnel detection for SSD, RetinaNet, YOLOv5, YOLOv7, YOLOv8, and Improved-YOLOv8 methods by more than 1% post-training. The Improved-YOLOv8 model trained on hybrid datasets reached AP_{50}, AP_{75}, and AP_{50-95} scores of 92.0%, 60.0%, and 54.1%, respectively, outperforming five comparative models. Compared to YOLOv8 trained on real datasets, the Improved-YOLOv8 notably improved subsidence funnel detection by 2.2%, 2.8%, and 2.0% in AP_{50}, AP_{75}, and AP_{50-95}.

Further experiments and analysis indicate that the Improved-YOLOv8 model trained on hybrid datasets demonstrates satisfactory generalization across diverse land surface types and other satellite datasets. Additionally, the model's detection accuracy is affected by temporal decorrelation, necessitating the selection of an appropriate temporal baseline for optimal detection performance.

This approach demonstrated high-precision automatic detection of subsidence funnels in mining regions. Specifically trained on mixed InSAR data, the Improved-YOLOv8 model performed robustly in detecting subsidence funnels across large-scale regions in Shanxi. Future applications aim to extend this method to nationwide subsidence funnel detection.

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